先进计算与数字工程研究中心

PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 03 – Generative Classifiers and Bayesian Decisions

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Three periods, one connected calculation

Time	Question and tools	What you will produce
0–45 min	Alarm counts → Bayes posterior → expected loss	A justified release/review rule
45–90 min	Labelled data → parameter estimates → Naive Bayes	A complete message prediction
90–135 min	Continuous data → Gaussian scores → class boundary	A fitted classifier and a worked decision

Learning target. Explain a probability, calculate it, and state the assumption that makes the calculation valid. Short pair exercises are part of each period.

An alarm is evidence, not a diagnosis

Factory model. In 10,000 parts, 2% are defective. The sensor alarms on 90% of defective parts and 5% of sound parts.

Object	Meaning	Notation
Unknown state	Defective or sound	$Y \in \{D, S\}$
Observed feature	Alarm or no alarm	$X \in \{A, \neg A\}$
Chosen action	Release, quarantine, review	a

Predict before calculating. Is $P(D \mid A)$ 90%, 50%, or below 50%? Explain which population your denominator counts.

Count all four outcomes before taking a ratio

True state	Alarm A	No alarm $\neg A$	Total
Defective D	$200 \times .90 = 180$	20	200
Sound S	$9,800 \times .05 = 490$	9,310	9,800
Total	670	9,330	10,000

$$P(A \mid D) = \frac{180}{200} = .90, \qquad P(D \mid A) = \frac{180}{670} \approx .269.$$

Interpretation. The same numerator appears in two different fractions. The second denominator includes every explanation of an alarm, including false alarms.

Count all four outcomes before taking a ratio

Inspection of 10,000 parts (teaching counts)

180 Defective + alarm	**490** Sound + alarm
20 Defective + no alarm	**9,310** Sound + no alarm

A joint table contains both forward and inverse questions

Probability	Alarm	No alarm	Row total
Defective	.018	.002	.020
Sound	.049	.931	.980
Column total	.067	.933	1

$$P(D, A) = P(A \mid D)P(D), \qquad P(D \mid A) = P(D, A)/P(A).$$

Read a row when conditioning on the state and a column when conditioning on the observation. Summing removes one variable; dividing by the conditioning event selects its population. A probability table must have nonnegative cells and total mass 1.

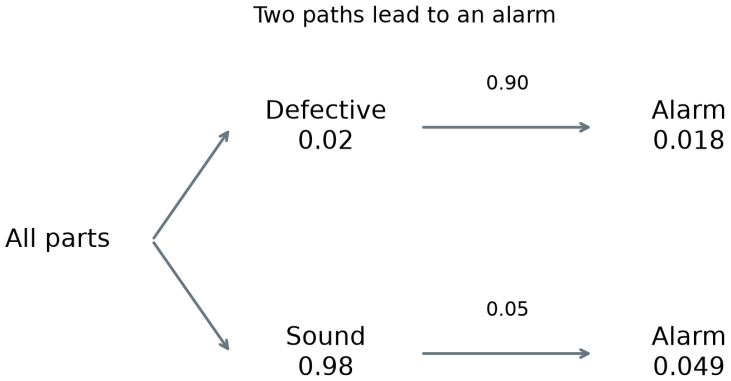
Bayes keeps every competing explanation

$$P(Y = k \mid x) = \frac{p(x \mid Y = k)\pi_k}{\sum_j p(x \mid Y = j)\pi_j}, \qquad \pi_k = P(Y = k).$$

Term	Meaning in the factory
Prior $\pi_D = .02$	Fraction defective before reading the sensor
Likelihood $P(A \mid D) = .90$	Alarm frequency among defective parts
Evidence $P(A) = .018 + .049$	Alarm frequency from all possible states
Posterior $P(D \mid A) = .018/.067$	Defect fraction after restricting to alarms

Multiply within one explanation; add across alternative explanations; normalize to obtain class probabilities.

Bayes keeps every competing explanation



Normalization changes the scale, not the winning class

Three class weights for one observation are $w = (.12, .09, .04)$.

$$Z = \sum_k w_k = .25, \qquad P(Y \mid x) = w/Z = (.48, .36, .16).$$

Operation	Result	Class choice
Compare w_k	Largest weight .12	A
Compare w_k/Z	Largest posterior .48	A
Multiply all weights by 100	$(12, 9, 4)$	A

Only a common positive factor may be removed. Multiplying one class alone changes the decision. Normalization is essential for expected-loss calculations, even when it is unnecessary for argmax.

Evidence multiplies the prior odds

$$\underbrace{\frac{P(D \mid A)}{P(S \mid A)}}_{\text{posterior odds}} = \underbrace{\frac{P(A \mid D)}{P(A \mid S)}}_{\text{likelihood ratio}} \underbrace{\frac{P(D)}{P(S)}}_{\text{prior odds}}.$$

$$18 \times \frac{1}{49} = \frac{18}{49}, \qquad P(D \mid A) = \frac{18}{18 + 49}.$$

Two observations. An alarm is 18 times more likely under a defect, so it is useful evidence. Sound parts were 49 times more common, so this evidence does not make a defect the most likely state. **Check.** If the likelihood ratio were 1, the observation would leave the prior odds unchanged.

Worked anomaly example: posterior exceeds the prior

Stated classroom rates. A rare event has prior .20. An unusual observation occurs with probability .60 when the event occurs and .10 otherwise.

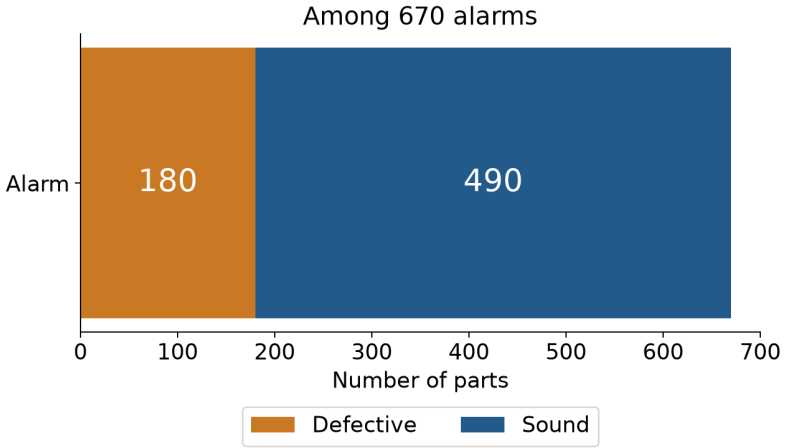
$$w_E = .20(.60) = .12, \quad w_N = .80(.10) = .08, \quad P(E \mid U) = .60.$$

Without the unusual observation,

$$P(E \mid \neg U) = \frac{.20(.40)}{.20(.40) + .80(.90)} = .10.$$

The observation changes the prior .20 to .60 or .10, depending on its value. These are illustrative rates adapted from the reference example. Both calculations use every competing explanation; a likelihood of .60 equals the first posterior here only by numerical coincidence.

Worked anomaly example: posterior exceeds the prior

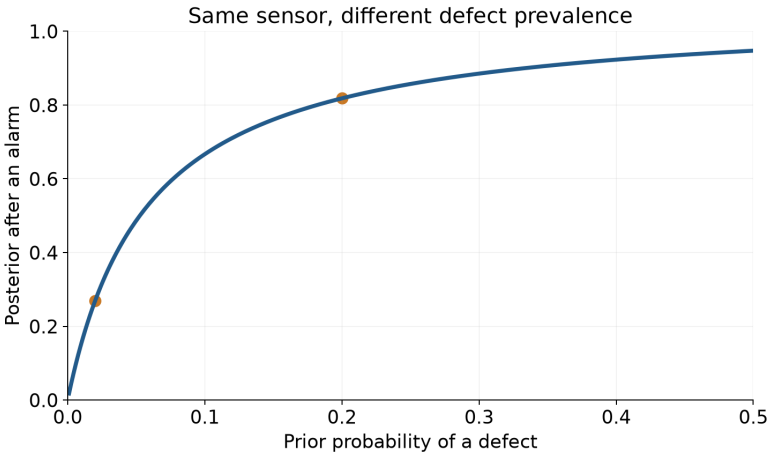


A green light and a new population give new posteriors

Situation	Calculation	Defect posterior
No alarm, 2% defects	$20/(20+9,310)$	.00214
Alarm, 2% defects	$.90(.02)/[.90(.02)+.05(.98)]$	.269
Alarm, 20% defects	$.90(.20)/[.90(.20)+.05(.80)]$	.818

What changed? The first comparison changes the observation. The second changes the class prior while keeping the sensor likelihoods fixed. A balanced training sample is a sampling choice; it does not establish a 50:50 class prior for every future population.

A green light and a new population give new posteriors



How much false-alarm rate can the alarm tolerate?

Keep defect prevalence .02 and sensitivity .90. Let $r = P(A \mid S)$ be the false-alarm rate. To make an alarm imply at least 50% defects, require

$$\frac{.02(.90)}{.02(.90) + .98r} \geq .5 \quad \Longrightarrow \quad r \leq \frac{.018}{.98} \approx .01837.$$

For an 80% defect posterior,

$$r \leq \frac{.018(1 - .8)}{.98(.8)} \approx .00459.$$

A detector for rare defects needs very few false alarms before a positive result becomes highly convincing. Improving sensitivity alone does not remove the influence of the large sound population.

Largest posterior minimizes zero-one error

Zero-one loss. A correct label costs 0; every wrong label costs 1.

$$R(k \mid x) = \sum_{j \neq k} P(Y = j \mid x) = 1 - P(Y = k \mid x).$$

Prediction after an alarm	Probability of error	Rule
Defective	.731	More errors
Sound	.269	Fewer errors

Decision. $\hat{y}(x) = \arg\max_k P(Y = k \mid x)$. This is a statement about the number of mistakes. A missed defect and a false alarm may have very different consequences, so a factory can reasonably need a different action rule.

The Bayes error is the probability mass that remains overlapped

Suppose X has three possible observations with joint class masses:

Observation	$P(X, A)$	$P(X, B)$	Best label
Left	.30	.05	A
Middle	.15	.20	B
Right	.05	.25	B

$$P(error)_{Bayes} = \sum_{x} \min\{P(x, A), P(x, B)\} = .05 + .15 + .05 = .25.$$

At each observation, the losing class contributes unavoidable zero-one error given only X. More training data can improve estimation; it does not by itself remove overlap already present in the true observation model.

Choose an action by its conditional expected loss

Action a	True D	True S	Risk after alarm
Quarantine Q	0	2	$2(49/67) = 1.463$
Release R	30	0	$30(18/67) = 8.060$

$$R(a \mid x) = \sum_j L(a, j) P(Y = j \mid x), \qquad a^*(x) = \arg\min_a R(a \mid x).$$

Read the table. Rows are chosen actions; columns are unknown true states. Each risk averages one row using the same posterior probabilities. Choose quarantine, even though sound is the more likely state. The action changed because a missed defect costs much more.

A general loss table need not have a zero diagonal

Let $p = P(D \mid x)$. Suppose quarantine costs 1 for a defect and 4 for a sound part; release costs 20 for a defect and .5 for a sound part.

$$R(Q) = p + 4(1-p) = 4 - 3p, \qquad R(R) = 20p + .5(1-p) = .5 + 19.5p.$$

$$R(Q) < R(R) \quad \Longrightarrow \quad p > \frac{3.5}{22.5} \approx .1556.$$

At $p = .10$, release risks 2.45 versus quarantine 3.70. At $p = .25$, quarantine risks 3.25 versus release 5.375. Correct actions can carry inspection or processing costs; the threshold should be derived from the complete stated table.

A threshold follows from the loss table

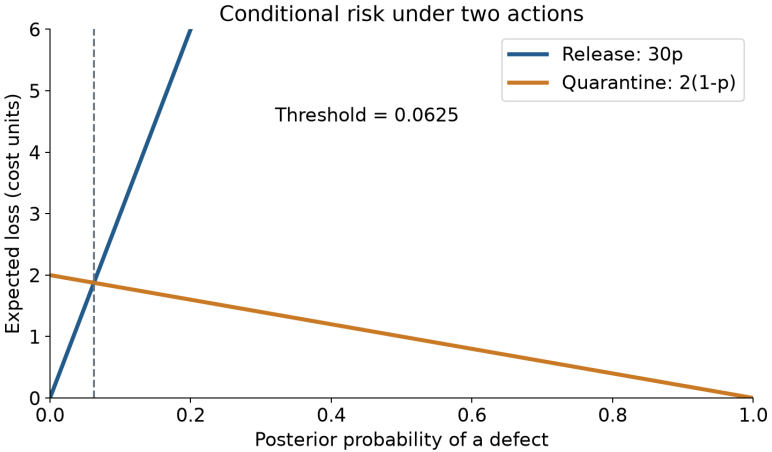
Let $p = P(D \mid x)$, with zero loss for correct actions. Quarantine when

$$c_{FA}(1-p) < c_M p \quad \Longleftrightarrow \quad p > \frac{c_{FA}}{c_M + c_{FA}}.$$

Miss cost c_M	False-alarm cost c_{FA}	Threshold	Effect
30	2	.0625	Intervene early
30	10	.25	Require stronger evidence
90	2	.0217	Avoid expensive misses

Factory answer. $.269 > .0625$, so quarantine. If costs change, recompute the threshold; the posterior remains an assessment of the state.

A threshold follows from the loss table



Likelihood-ratio testing expresses the same risk decision

With zero correct-action loss, quarantine when posterior odds satisfy

$$\frac{P(D \mid x)}{P(S \mid x)} > \frac{c_{FA}}{c_M}.$$

Bayes gives an equivalent likelihood-ratio rule:

$$\frac{p(x \mid D)}{p(x \mid S)} > \frac{c_{FA}}{c_M}\frac{\pi_S}{\pi_D}.$$

For factory priors $(.02, .98)$ and costs $(c_M, c_{FA}) = (30, 2)$, the threshold is $(2/30)49 \approx 3.267$. The alarm likelihood ratio is 18 and the no-alarm ratio is $.10/.95 \approx .1053$. Thus quarantine alarms and release non-alarms, matching the conditional-risk calculations.

Review creates an intermediate action region

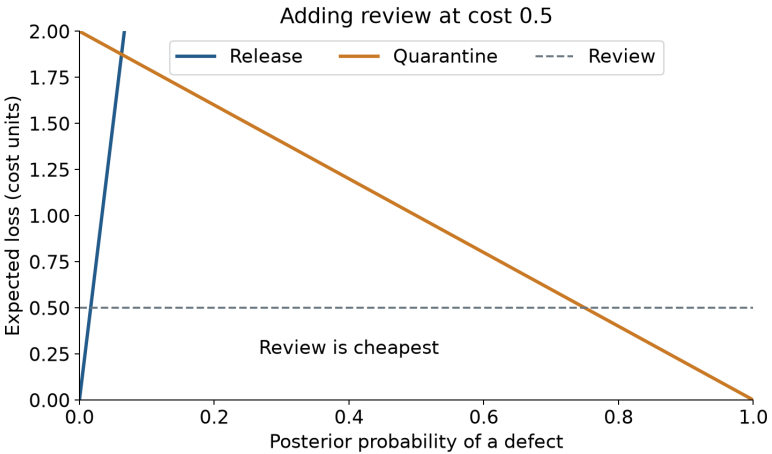
Add review with cost .5 in either state:

$$R(R \mid x) = 30p, \qquad R(Q \mid x) = 2(1-p), \qquad R(V \mid x) = .5.$$

Posterior region	Lowest-risk action	Why
$p < 1/60$	Release	$30p < .5$
$1/60 < p < .75$	Review	Both automatic actions cost more
$p > .75$	Quarantine	$2(1-p) < .5$

At the alarm posterior $p = .269$, review has the smallest risk. This example treats .5 as the total expected review loss; delays and reviewer errors would need to be included in a richer loss model.

Review creates an intermediate action region



Symmetric reject rules use the largest class posterior

Suppose every wrong class costs $L_M = 1$, correct predictions cost 0, and rejection costs $L_R = .20$ regardless of class.

$$R(k \mid x) = 1 - P(k \mid x), \qquad R(\textit{reject} \mid x) = .20.$$

Choose the best class only when $\max_k P(k \mid x) > .80$.

Posterior vector	Best class risk	Action
$(.85, .10, .05)$	.15	Predict first class
$(.60, .25, .15)$	.40	Reject

This simple largest-posterior threshold relies on equal misclassification costs and class-independent rejection loss. Unequal costs require comparing all action risks, as in the asymmetric factory model.

Pointwise choices determine the overall risk

$$\mathcal{R}(a) = E_X[R(a(X) \mid X)], \qquad \mathcal{R}^* = E_X[\min_a R(a \mid X)].$$

With only release/quarantine available, quarantine alarms and release non-alarms:

$$\mathcal{R} = \frac{490(2) + 20(30)}{10,000} = .158 \text{ per part.}$$

Quantity	Interpretation
Conditional risk	Average loss for this observed input
Expected risk	Average over all future inputs
Bayes risk	Best possible under the true distribution, loss, and available actions

Estimated probabilities can differ from truth, so fitting a Bayes rule does not automatically attain the true Bayes risk.

A threshold changes the counts of different mistakes

On a labelled evaluation set containing 20 defects and 80 sound parts:

Policy	Misses	False alarms	Total loss
Higher threshold	2	8	$30(2) + 2(8) = 76$
Lower threshold	1	16	$30(1) + 2(16) = 62$

Accuracy is 90% for the first policy and 83% for the second. Lowering the threshold catches one extra defect but quarantines eight extra sound parts. Under the stated costs, avoiding that missed defect is worth 30 and the additional false alarms cost 16. Compare consequences, not accuracy alone.

Worked transfer: keep or drop a course?

Adapted course-choice example. Prior $P(G) = .6$ for a good course. An engaging class has likelihood .8 under G and .1 under a poor course.

$$P(G \mid E) = \frac{.6(.8)}{.6(.8) + .4(.1)} = \frac{12}{13}.$$

Action	Good course	Poor course	Risk given engaging class
Keep	0	10	10/13
Drop	20	0	240/13

Choose keep. The units are teaching loss units. First infer the unknown course quality; then average the consequences of each action. An engaging class is one observation, not a direct measurement of total course quality.

The boring-class observation can reverse the course decision

Keep the reference course model: $P(G) = .6$, $P(E \mid G) = .8$, $P(E \mid \neg G) = .1$. This time the class is boring, so

$$P(G \mid \neg E) = \frac{.6(.2)}{.6(.2) + .4(.9)} = .25.$$

With keep-poor loss 10 and drop-good loss 20:

$$R(\mathit{keep}) = 10(.75) = 7.5, \qquad R(\mathit{drop}) = 20(.25) = 5.$$

Drop has lower expected loss. A good course can give a boring class; one observation does not determine quality with certainty. The decision is conditional on the observation and the stated costs, so a different lesson observation can reasonably change the action.

More classes: compare weights, then compare actions

Three competing classes

Class	Prior	Likelihood	Weight
A	.40	.30	.12
B	.30	.30	.09
C	.30	2/15	.04

$$P(Y \mid x) = (.48, .36, .16).$$

Largest posterior selects A.

A separate loss example

Action	A	B	C
Choose A	0	4	8
Choose B	2	0	3
Choose C	5	1	0

For $P(Y \mid x) = (.60, .25, .15)$, risks are $(2.20, 1.65, 3.25)$; choose B.

The two tables use different stated posteriors. Class inference and action optimization are related calculations with different objectives.

Multiclass risk is a dot product for each action

Use posterior $(.50, .30, .20)$ and this action-by-truth loss table:

Action	A	B	C	Conditional risk
Choose A	0	2	8	$0 + .6 + 1.6 = 2.2$
Choose B	3	0	2	$1.5 + 0 + .4 = 1.9$
Choose C	6	1	0	$3 + .3 + 0 = 3.3$

Class A is most likely, but action B has minimum risk. Matrix notation is $r = Lp$ when rows are actions and columns are true classes. The three posterior probabilities are reused for every row; they must not be reordered to match the chosen action.

Pair exercise: learn rates and choose an action

New labelled records. 10 defective parts include 8 alarms. 90 sound parts include 18 alarms. Use raw frequency estimates in this exercise.

1. Estimate $P(D)$, $P(A \mid D)$, and $P(A \mid S)$.
2. A new part alarms. Calculate both class weights and $P(D \mid A)$.
3. With miss cost 30 and false-alarm cost 2, compare release and quarantine.
4. Add review cost .5 in either state. Which action now wins?

Explain to your partner. State the denominator at each step. Do not stop after choosing the most likely class. Keep exact fractions until the final comparison.

Exercise: infer the positive-result population before acting

A screening model has prevalence .05, sensitivity .80, and false-positive rate .10. Let $+$ denote a positive result.

1. Out of 2,000 observations, count true positives and false positives.
2. Compute $P(D \mid +)$ and the complementary class posterior.
3. A miss costs 12; unnecessary intervention costs 3. Choose the lower-risk action.
4. Add a review action costing .4 in both states. Compare all three risks.

Write the relevant population beside every denominator. The sensor rate among defects and the defect fraction among positives are different questions. Keep the posterior unchanged when adding an action.

Solution: one extra action changes the minimum

In 2,000 observations, there are 100 defects and 1,900 sound parts. Positive counts are 80 and 190.

$$P(D \mid +) = \frac{80}{270} = \frac{8}{27}, \qquad P(S \mid +) = \frac{19}{27}.$$

Action	Risk	Comparison
Intervene	$3(19/27) = 19/9$	Lower without review
Release	$12(8/27) = 32/9$	Higher
Review	.4	Lowest when available

The state probability is about .296, even though intervention is preferable to release. Adding review changes the best action to review; it does not change how the positive observation updates the state.

Sensitivity to the stated loss model is a useful final check

Use the previous posterior $p = 8/27$. Correct-action losses remain zero, but change the missed-defect cost.

Miss cost	Release risk	Intervention risk	Best of two actions
2	$16/27 = .593$	$19/9 = 2.111$	Release
12	$32/9 = 3.556$	$19/9 = 2.111$	Intervene
60	$160/9 = 17.778$	$19/9 = 2.111$	Intervene

For this fixed posterior, the two actions tie when $c_M p = 3(1-p)$, giving $c_M = 57/8 = 7.125$. State uncertainty and cost uncertainty are different inputs to the decision; vary one while holding the other fixed.

Solution: probability and action are separate steps

$$\hat{\pi}_D = .10, \quad \hat{P}(A \mid D) = .80, \quad \hat{P}(A \mid S) = .20.$$

$$w_D = .08, \quad w_S = .18, \quad P(D \mid A) = \frac{.08}{.26} = \frac{4}{13}.$$

Action	Calculation	Conditional risk
Release	$30(4/13)$	$120/13 \approx 9.231$
Quarantine	$2(9/13)$	$18/13 \approx 1.385$
Review	.5	.5

Without review: quarantine. With review: review. **What changed?** The available actions changed. The posterior $4/13$ did not. In the next period, we study how frequency estimates can become unreliable when the training sample is small.

A distribution family still needs parameter values

Object	Example	What varies?
Observation X	A success or failure	Random outcome
Parameter θ	Success probability	Unknown model quantity
Observed data $\mathcal{D}$	Seven successes, three failures	Fixed once recorded

$$L(\theta) = p(\mathcal{D} \mid \theta) = \theta^7(1-\theta)^3.$$

Likelihood. Keep these data fixed and compare candidate parameters. $L(\theta)$ is not automatically a normalized density over θ. **Two approaches.** MLE selects the parameter explaining the data best. Bayesian estimation updates a distribution over plausible parameter values.

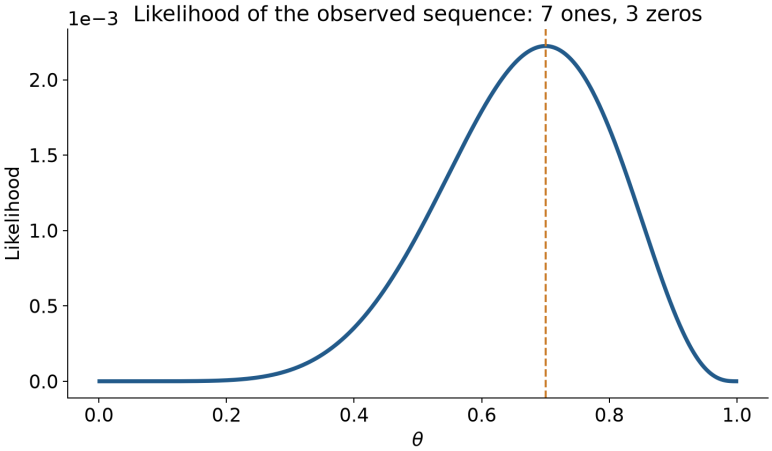
Compare candidate parameters with the same fixed data

The observed sequence contains 7 successes and 3 failures. Its likelihood is $L(\theta) = \theta^7(1-\theta)^3$.

Candidate θ	Sequence likelihood	Explanation
.30	.0000750	Seven successes are hard to explain
.50	.0009766	Symmetric model
.70	.0022236	Best of these candidates
.90	.0004783	Three failures are hard to explain

These are probabilities of the same recorded sequence under different models, not probabilities assigned to the parameters. They need not sum to 1 across candidate values. Maximization asks which parameter makes these observations most plausible.

Compare candidate parameters with the same fixed data



A count likelihood differs by a parameter-independent factor

For one specific sequence with seven successes, $L_{seq}(\theta) = \theta^7(1-\theta)^3$. If only the success count is observed,

$$P(S = 7 \mid \theta) = \binom{10}{7}\theta^7(1-\theta)^3 = 120 L_{seq}(\theta).$$

At $\theta = .7$, the sequence probability is about .002224 and the count probability is about .266828. There are 120 sequences with that same count.

Both likelihoods maximize at .7 because 120 is independent of θ. The observation definition changes the probability being computed. When comparing classes or parameters, remove a factor only after checking that it is common to every competitor.

Derive the Bernoulli maximum-likelihood estimate

For s successes in n independent trials with common probability θ:

$$\ell(\theta) = s \log \theta + (n - s) \log(1 - \theta),$$

$$\ell'(\theta) = \frac{s}{\theta} - \frac{n - s}{1 - \theta} = 0 \quad \Longrightarrow \quad \hat{\theta}_{ML} = \frac{s}{n}.$$

$$\ell''(\theta) = -\frac{s}{\theta^2} - \frac{n - s}{(1 - \theta)^2} < 0 \quad (0 < s < n).$$

Example. Seven successes and three failures give .7. All failures or all successes give boundary estimates 0 or 1. Logarithms preserve the maximizer, turn products into sums, and avoid multiplying many tiny probabilities. The result relies on independent trials with one shared probability.

A labelled joint likelihood separates prior and sensor rates

From 100 labelled parts, 10 are defective: 8 alarm and 2 do not. Of 90 sound parts, 18 alarm and 72 do not. Let $\pi = P(D)$, $s = P(A \mid D)$, and $r = P(A \mid S)$.

$$L(\pi, s, r) \propto \pi^{10}(1-\pi)^{90} s^{8}(1-s)^{2} r^{18}(1-r)^{72}.$$

The log likelihood is a sum of three Bernoulli fitting problems, giving

$$\hat{\pi} = .10, \qquad \hat{s} = .80, \qquad \hat{r} = .20.$$

Labels attach each observation to the correct class-conditional term. A supervised generative model learns class frequency and feature distributions separately because the complete labelled likelihood factors this way.

More repeated evidence sharpens the likelihood around the estimate

For an interior Bernoulli MLE $\hat{\theta} = s/n$, the negative curvature of the log likelihood is

$$-\ell''(\hat{\theta}) = \frac{n}{\hat{\theta}(1-\hat{\theta})}.$$

Data with same fraction	MLE	Approximate standard error
7 successes of 10	.70	$\sqrt{.21/10} = .145$
70 successes of 100	.70	$\sqrt{.21/100} = .046$

The estimate is unchanged, but its sampling precision improves. These standard errors use a large-sample interior approximation; they are not guaranteed exact uncertainty intervals for small or boundary-valued samples. A point estimate alone hides the amount of evidence behind it.

Beta updating records evidence and uncertainty

$$p(\theta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}, \qquad \theta \mid \mathcal{D} \sim \mathrm{Beta}(\alpha+s, \beta+n-s).$$

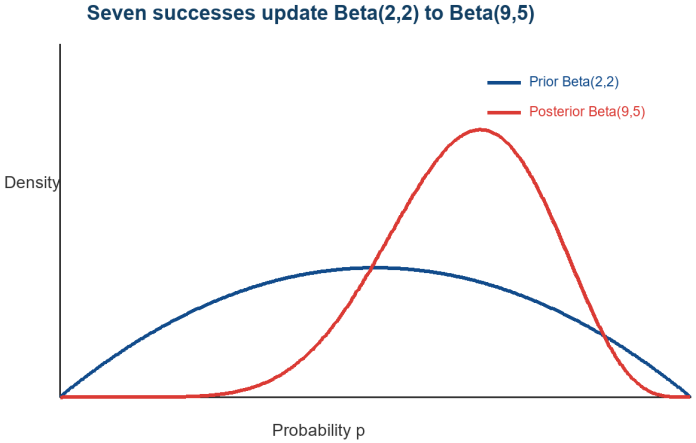
Estimate with prior Beta$(2,2)$ and data $(7,3)$	Formula	Value
Maximum likelihood	$7/10$	.700
Posterior mean	$9/(9+5)$	.643
Posterior mode	$(9-1)/(14-2)$	.667

The posterior is Beta$(9,5)$, not merely one adjusted number. Its mean averages plausible parameters; its mode selects the most probable parameter. The displayed interior mode formula requires both posterior shape parameters to exceed 1.

Sequential check. Add counts $(3,1)$, then $(4,2)$. The posterior matches a one-batch update:

$$\mathrm{Beta}(2,2) \to \mathrm{Beta}(5,3) \to \mathrm{Beta}(9,5).$$

Beta updating records evidence and uncertainty



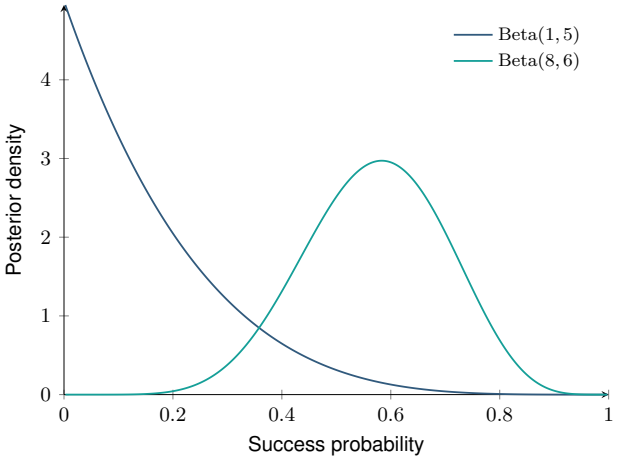
A strong prior changes the interpretation of four failures

Observe zero successes in four trials. Compare two explicitly stated priors:

Prior	Posterior	Predictive success probability
$\text{Beta}(1,1)$	$\text{Beta}(1,5)$	$1/6 = .167$
$\text{Beta}(8,2)$	$\text{Beta}(8,6)$	$8/14 = .571$

The second prior initially favors high success probability and has greater effective count. Four failures move it downward, but do not outweigh all prior information. Smoothing is an assumption about plausible parameters, not a universal add-one rule. State the prior and compare its influence with the amount of observed evidence.

A strong prior changes the interpretation of four failures



A posterior mode at zero does not imply zero predictive probability

A uniform Beta$(1,1)$ prior and four failures produce

$$p(\theta \mid \mathcal{D}) \propto (1-\theta)^4, \qquad \theta \mid \mathcal{D} \sim \mathit{Beta}(1,5).$$

The density decreases from $\theta = 0$, so its mode is at the boundary 0. Yet

$$E[\theta \mid \mathcal{D}] = \frac{1}{6}, \qquad P(X_{new} = 1 \mid \mathcal{D}) = \frac{1}{6}.$$

The mode selects one parameter value; prediction averages over all plausible parameter values. Do not use the interior Beta mode formula without checking its assumptions. An estimated optimum of zero and nonzero uncertainty-supported prediction can both be correct under the same posterior.

Prediction and Laplace smoothing average uncertainty

$$P(X_{new} = 1 \mid \mathcal{D}) = \int \theta p(\theta \mid \mathcal{D}) d\theta = \frac{\alpha + s}{\alpha + \beta + n}.$$

$$E[\theta \mid \mathcal{D}] = \frac{n}{n + \alpha + \beta} \frac{s}{n} + \frac{\alpha + \beta}{n + \alpha + \beta} \frac{\alpha}{\alpha + \beta}.$$

Prior	Data	Predictive success probability
Beta$(2, 2)$	7 of 10	$9/14 = .643$
Beta$(2, 2)$	70 of 100	$72/104 = .692$
Beta$(1, 1)$	0 of 4	$1/6$, rather than 0

Add-one smoothing is the posterior-mean prediction for a uniform Beta prior. Do not call this the MAP estimate: under the same uniform prior, MAP equals MLE.

Parameter uncertainty differs from the randomness of the next outcome

For the posterior Beta$(9, 5)$,

$$E[\theta] = 9/14 = .643, \quad Var(\theta) = \frac{9(5)}{14^2(15)} \approx .01531.$$

The parameter standard deviation is about .124. The next binary outcome has predictive variance

$$Var(X_{new} \mid \mathcal{D}) = \frac{9}{14}\left(1 - \frac{9}{14}\right) \approx .22959.$$

These variances describe different objects: uncertainty about an unknown success probability versus randomness of one future zero/one outcome. More data can narrow the parameter posterior while individual outcomes remain random. A precise estimated probability is not a promise about the next trial.

Categorical maximum likelihood needs the sum-to-one constraint

Counts for three categories are $(2, 3, 5)$, with probabilities $\theta_1, \theta_2, \theta_3$.

$$\ell = 2 \log \theta_1 + 3 \log \theta_2 + 5 \log \theta_3, \qquad \sum_j \theta_j = 1.$$

Use $\mathcal{L} = \sum_j n_j \log \theta_j + \lambda(1 - \sum_j \theta_j)$. Stationarity gives $n_j/\theta_j = \lambda$. Summing $\theta_j = n_j/\lambda$ gives $\lambda = 10$, hence

$$\hat{\theta} = (.2, .3, .5).$$

The constraint makes the separate counts share a denominator. Maximizing each term independently without normalization would not define a valid categorical distribution. The result generalizes the Bernoulli sample proportion.

More outcomes require a normalized probability vector

For M mutually exclusive outcomes with counts $n_1, \ldots, n_M$:

$$\hat{\theta}_{j,ML} = \frac{n_j}{n}, \qquad E[\theta_j \mid \mathcal{D}] = \frac{n_j + \alpha_j}{n + \sum_v \alpha_v}.$$

Three observed categories	First	Second	Third
Counts	3	1	0
MLE probabilities	$3/4$	$1/4$	0
Dirichlet$(1,1,1)$ posterior mean	$4/7$	$2/7$	$1/7$

The posterior is Dirichlet$(4,2,1)$; the estimates sum to 1. A categorical feature with M values adds M counts to its denominator. A word-count model instead uses the total number of tokens in that class.

Dirichlet smoothing can express a nonuniform prior

Observed category counts are $(3, 1, 0)$. Start with Dirichlet prior parameters $\alpha = (1, 2, 1)$.

$$\theta \mid \mathcal{D} \sim \textit{Dirichlet}(4, 3, 1), \qquad E[\theta \mid \mathcal{D}] = (4/8, 3/8, 1/8).$$

Prior	Prior mean	Posterior mean
Uniform $(1, 1, 1)$	$(1/3, 1/3, 1/3)$	$(4/7, 2/7, 1/7)$
Nonuniform $(1, 2, 1)$	$(1/4, 1/2, 1/4)$	$(1/2, 3/8, 1/8)$

Both assign nonzero probability to the unobserved third category. The different prior weight on the second category changes the final estimate, even though the observed counts are identical.

Conditional independence makes a large table manageable

$$p(x_1, \ldots, x_d \mid Y = k) \approx \prod_{j=1}^{d} p(x_j \mid Y = k).$$

Model for d binary features	Class-conditional parameters per class
Full table of all feature patterns	$2^d - 1$
One Bernoulli probability per feature	d

For $d = 20$, this is 1,048,575 versus 20. The simplification replaces interactions with an independence assumption **within each class**. **Training.** Estimate class priors from class counts; estimate each feature distribution using only samples from that class. Features need not be independent after the classes are mixed.

A dependent feature pair can fool the naive product

Within one class, suppose two binary features have this true joint distribution:

	$X_2 = 0$	$X_2 = 1$
$X_1 = 0$	.40	.10
$X_1 = 1$	.10	.40

Both marginal probabilities of 1 are .50. A naive product therefore assigns .25 to $(1, 1)$, while the true joint probability is .40.

Conditional independence is a model assumption, not a consequence of splitting the training data by class. The joint table contains a relationship omitted by the product. A compact model needs less data but can misrepresent evidence combinations.

Bernoulli features: present and absent both count

Training messages	Spam	Ordinary
Class count n_k	4	6
Messages containing free	3	1
Messages containing meeting	0	4

$$\phi_{kj} = \frac{n_{kj} + 1}{n_k + 2}, \qquad p(x \mid k) = \prod_j \phi_{kj}^{x_j} (1 - \phi_{kj})^{1 - x_j}.$$

Representation. $x = (1, 0)$ means free is present and meeting is absent. Repeating free twice still gives the same binary value 1. For spam: $\phi = (4/6, 1/6)$; for ordinary mail: $(2/8, 5/8)$. Priors remain .4 and .6 in this stated example.

Bernoulli naive Bayes gives a linear log-odds score

For two classes, expanding presence and absence terms yields

$$\log \frac{P(S \mid x)}{P(O \mid x)} = b + \sum_j x_j v_j,$$

$$v_j = \log \frac{\phi_{Sj}(1-\phi_{Oj})}{\phi_{Oj}(1-\phi_{Sj})}, \quad b = \log \frac{\pi_S}{\pi_O} + \sum_j \log \frac{1-\phi_{Sj}}{1-\phi_{Oj}}.$$

With the fitted free/meeting probabilities, $v_{free} = \log 6$, $v_{meeting} = \log(.12)$, and $b \approx -.4179$. Input $(1, 0)$ gives score about 1.3739, implying spam posterior .798.

A generative independence model can therefore produce a linear score in binary features. Its coefficients come from probability estimates rather than direct boundary fitting.

A complete Bernoulli message prediction

New message: free present, meeting absent, $x = (1, 0)$.

$$w_S = .4\left(\frac{4}{6}\right)\left(1 - \frac{1}{6}\right) = \frac{2}{9} \approx .2222,$$

$$w_O = .6\left(\frac{2}{8}\right)\left(1 - \frac{5}{8}\right) = .05625,$$

$$P(S \mid x) = \frac{w_S}{w_S + w_O} = \frac{320}{401} \approx .798.$$

Read the factors. Prior $\times$ probability of free $\times$ probability of no meeting. Both classes explain the same observation. **Question.** If meeting also appears, which factors must change? Replace each absence factor by its presence probability. Do not alter the class prior merely because this new message has changed.

The same fitted model can reverse its prediction when a word appears

The fitted Bernoulli model has $\pi = (.4, .6)$, $\phi_S = (2/3, 1/6)$, and $\phi_O = (1/4, 5/8)$. For input (*free*, *meeting*) $= (1, 1)$:

$$w_S = .4(2/3)(1/6) = 2/45, \qquad w_O = .6(1/4)(5/8) = 3/32.$$

$$P(S \mid x) = \frac{2/45}{2/45 + 3/32} = \frac{64}{199} \approx .322.$$

The earlier input $(1, 0)$ gave .798. Adding meeting replaces an absence factor by a presence factor in each class. It adds evidence favoring ordinary mail; the trained parameters and class priors remain unchanged.

Categorical features: borrow the account example

Reference example. Predict whether an account is genuine from posting density, friend density, and a real profile photo. Density takes s, m, l; photo takes yes/no.

For new input (m, m, yes)	Genuine	Not genuine
Prior	$7/10$	$3/10$
Posting likelihood	$2/5$	$1/3$
Friend likelihood	$1/2$	$1/6$
Photo likelihood	$5/9$	$3/5$

$$w_G = 7/90, \quad w_N = 1/100, \quad P(G \mid x) = \frac{70}{79} \approx .886.$$

The source uses raw class priors and smoothed feature probabilities; keep these choices explicit. Categories are alternatives, not numeric distances between s, m, l.

Multinomial features count tokens, not messages

Training token counts	free	meeting	project	Total
Spam	6	0	2	8
Ordinary	1	5	4	10

$$\phi_{kj} = \frac{n_{kj} + 1}{\sum_v n_{kv} + V}, \qquad V = 3.$$

New message. free free project has count vector $x = (2, 0, 1)$.

$$w_S = .4(7/11)^2(3/11) = .04418, \qquad w_O = .6(2/13)^2(5/13) = .005462.$$

Result. $P(S \mid x) \approx .890$. Repeated free contributes twice; absent meeting contributes no token factor. The class-independent multinomial coefficient cancels when comparing classes.

A token model conditions on the observed message length

For count vector $x = (2, 0, 1)$, the length is $N = 3$. A multinomial model gives

$$p(x \mid k, N) = \frac{N!}{\prod_j x_j!} \prod_j \phi_{kj}^{x_j} = 3\phi_{k,free}^2 \phi_{k,project}.$$

The factor 3 counts the possible token orders sharing these counts. It is common across classes and cancels in class posterior ratios.

This model treats the observed length as given. If message length itself differs across classes and should be evidence, add an explicit $p(N \mid k)$ model. Do not silently claim that token probabilities already describe every aspect of the document.

Count features also add linear evidence in log odds

For a fixed vocabulary and observed count vector,

$$\log \frac{P(S \mid x)}{P(O \mid x)} = \log \frac{\pi_S}{\pi_O} + \sum_j x_j \log \frac{\phi_{Sj}}{\phi_{Oj}}.$$

In the free/meeting/project model:

Term	Ratio	Log contribution for $(2, 0, 1)$
Prior	$.4/.6$	$\log(2/3)$
free	$(7/11)/(2/13)$	$2\log(91/22)$
project	$(3/11)/(5/13)$	$\log(39/55)$

Their sum is about 2.0904, giving posterior .890. Repeated free contributes twice; an absent token has exponent zero, unlike an absent Bernoulli word feature.

Log scores connect the formula to the program

$$s_k = \log \pi_k + \sum_j x_j \log \phi_{kj}, \qquad P(k \mid x) = \frac{e^{s_k - m}}{\sum_j e^{s_j - m}}, \quad m = \max_j s_j.$$

```
import numpy as np
counts = np.array([[6.,0.,2.], [1.,5.,4.]])
phi = (counts+1)/(counts.sum(axis=1, keepdims=True)+3)
x = np.array([2.,0.,1.])
s = np.log([.4,.6]) + np.log(phi) @ x
q = np.exp(s-s.max())
print(q/q.sum())  # approximately [.890, .110]
```

Rows are classes; columns are words. Subtracting a common maximum preserves score differences and the normalized posterior. Predict by argmax when probabilities are not needed.

What independence and representation can lose

Example	Model effect	What was lost?
Copy one clue into two columns	Odds 2 : 1 become 4 : 1	Dependence is counted twice
“Dog bites man” versus “Man bites dog”	Identical count vectors	Word order and roles
20 binary clues with little training data	Few parameters help estimation	Interactions are omitted

Conditional, not marginal. With equal class priors and independent feature probabilities .9 in A and .1 in B, $P(X_1 = 1) = P(X_2 = 1) = .5$, but $P(X_1 = X_2 = 1) = .41 \neq .25$. A useful classification rule can still give overconfident probabilities. Richer features help only if their added information is supported by data.

Choose a distribution from the observation definition

Observation	Model	Smoothed or fitted quantity
Present/absent word	Bernoulli	$(n_{kj} + 1)/(n_k + 2)$
One of M categories	Categorical	$(n_{kjm} + 1)/(n_k + M)$
Counts over V words	Multinomial	$(n_{kj} + 1)/(\text{class tokens} + V)$
Continuous feature	Gaussian	Class mean and variance

Pair exercise. A class has 4 records, with category counts $(3, 1, 0)$. Give the smoothed probabilities. Then explain why using denominator 6 would be wrong. **Answer.** $(4/7, 2/7, 1/7)$; three possible categories require three added counts. A continuous measurement is not automatically Gaussian; the observed shape still matters.

A Gaussian mean can also be updated Bayesianly

Known observation variance. $x_i \mid \mu \sim \mathcal{N}(\mu, \sigma^2)$ and $\mu \sim \mathcal{N}(\mu_0, \tau_0^2)$.

$$\tau_n^{-2} = \tau_0^{-2} + n\sigma^{-2}, \qquad \mu_n = \tau_n^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{x}}{\sigma^2} \right).$$

Numerical example. Prior $\mu_0 = 0$, $\tau_0^2 = 4$; two readings with mean $\bar{x} = 3$ and known $\sigma^2 = 1$.

$$\tau_n^2 = \frac{1}{.25 + 2} = \frac{4}{9}, \qquad \mu_n = \frac{4}{9}(6) = \frac{8}{3}.$$

The data pull the mean from 0 toward 3. τ_n^2 measures uncertainty about the unknown mean; it is distinct from the observation noise variance σ^2. More independent readings add precision.

A naive Bayes model can combine different feature types

Observe a binary alarm flag, a material category M, and continuous vibration $v = 2$. Priors are $(.6, .4)$.

Feature model	Class A	Class B
$P(flag = 1 \mid k)$	.8	.2
$P(material = M \mid k)$	.5	.25
$v \mid k$	$\mathcal{N}(2, 1)$	$\mathcal{N}(0, 1)$

The Gaussian likelihood ratio at $v = 2$ is e^2. Combining the prior and feature ratios gives

$$\frac{P(A \mid x)}{P(B \mid x)} = (.6/.4)(.8/.2)(.5/.25)e^2 = 12e^2.$$

Thus $P(A \mid x) \approx .98885$. Each feature uses the distribution matching its observation type. Multiplication still assumes conditional independence within a class.

Exercise: select the likelihood family and estimate its parameters

A class has 5 labelled records. A binary flag appears in 1 record; a three-valued category has counts $(2, 3, 0)$; token counts over a vocabulary of 3 words are $(8, 1, 1)$.

1. Give the add-one estimate for flag presence.
2. Give all three smoothed category probabilities.
3. Give the three smoothed token probabilities.
4. Explain the three different denominators, even though all observations belong to the same class.

Do not use 5 as the denominator of the token model. Class records, category observations, and token occurrences are different counting units. Validate that each probability vector sums to 1.

Solution: the counting unit determines the denominator

Observation model	Smoothed estimate	Denominator meaning
Binary presence	$(1+1)/(5+2) = 2/7$	5 records plus 2 binary outcomes
Three categories	$(3/8, 4/8, 1/8)$	5 records plus 3 categories
Token counts	$(9/13, 2/13, 2/13)$	10 token occurrences plus 3 words

All entries are valid probabilities; the categorical and token vectors each sum to 1. The class prior would use class-record frequency among all classes, a fourth counting question. Choosing a distribution is part of the problem definition, not a detail postponed until coding.

One training table produces priors and feature models

From labels to a fitted generative classifier

1. Count labels to estimate $\pi_k = P(Y = k)$.
2. Split training observations by class; define each feature's sample space.
3. Estimate class-conditional parameters, using the stated smoothing or MLE rule.
4. For a new x, compute $s_k = \log \pi_k + \log p(x \mid k)$.
5. Normalize if probabilities are required, then apply the specified loss table.

Check your understanding. For the message example, class priors count messages and word probabilities count tokens. For a sensor reading, the feature model will describe a continuous density. The classifier pipeline stays the same; its observation model changes.

Continuous measurements need densities, not point masses

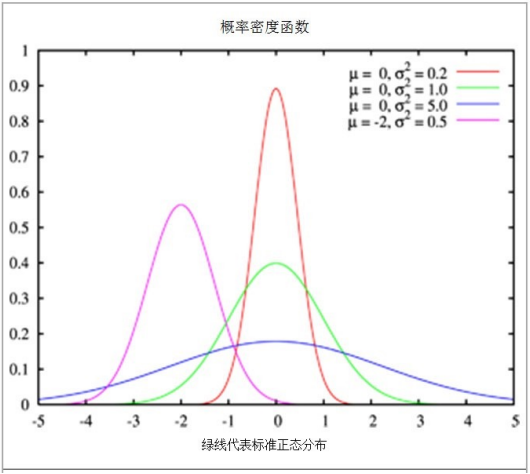
For $X \sim \mathcal{N}(\mu, \sigma^2)$,

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right], \qquad P(a < X < b) = \int_a^b p(x)dx.$$

Quantity for $\mathcal{N}(0, .01)$	Value	Meaning
Density at 0	3.989	Local concentration per input unit
Probability on $[-.1, .1]$	.683	Area under the density
Probability at exactly 0	0	A single point has zero width

Class posteriors compare class-weighted densities at the same input. A density height is not itself a class probability and can exceed 1.

Continuous measurements need densities, not point masses



Fit a one-dimensional Gaussian mean and variance from the likelihood

For independent $x_i \sim \mathcal{N}(\mu, \sigma^2)$,

$$\ell = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_i (x_i - \mu)^2.$$

Setting the mean derivative to zero gives $\sum_i (x_i - \mu) = 0$, so $\hat{\mu} = \bar{x}$. The variance derivative then gives

$$\hat{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \bar{x})^2.$$

For data $(2, 4, 6)$, $\hat{\mu} = 4$, squared deviations are $(4, 0, 4)$, and $\hat{\sigma}^2 = 8/3$. The mean locates the center; the variance fits the within-model spread. The MLE denominator is n, consistent with the objective being maximized.

A common unit change cancels from Gaussian class comparisons

Let $z = cx$ with $c > 0$. A one-dimensional density transforms as

$$p_Z(z \mid k) = \frac{1}{c} p_X(z/c \mid k), \quad \mu_{Z,k} = c\mu_{X,k}, \quad \sigma^2_{Z,k} = c^2 \sigma^2_{X,k}.$$

If a measurement is converted from meters to centimeters, $c = 100$ and the density height is divided by 100. Both class densities acquire the same factor, so

$$\frac{p_Z(z \mid A)\pi_A}{p_Z(z \mid B)\pi_B} = \frac{p_X(x \mid A)\pi_A}{p_X(x \mid B)\pi_B}.$$

Posterior and label remain unchanged when data and model parameters are transformed consistently. Density has units; a class probability does not depend on the chosen measurement unit.

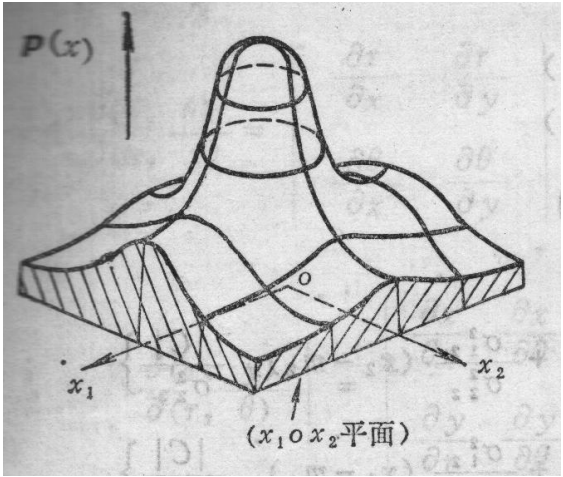
A covariance matrix describes variation and correlation

$$\mu = E[x], \qquad \Sigma = E[(x-\mu)(x-\mu)^T], \qquad x = (x_1, x_2)^T.$$

Entry	Interpretation	Example
Σ_{11}, Σ_{22}	Variances of the two measurements	$4, 1$
$\Sigma_{12} = \Sigma_{21}$	Joint deviations around the means	Positive: often increase together
$\rho = \Sigma_{12}/\sqrt{\Sigma_{11}\Sigma_{22}}$	Unit-free correlation	$\Sigma_{12} = 1 \Rightarrow \rho = .5$

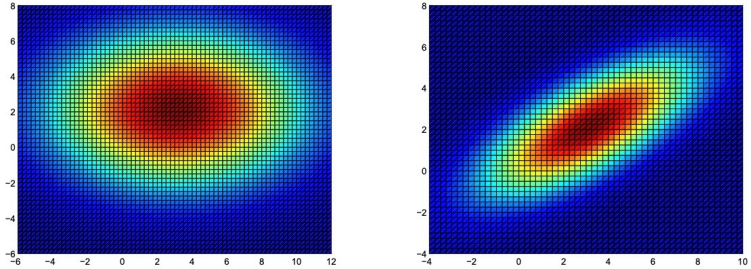
Example. $\Sigma = \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix}$ is symmetric and positive definite. Its contour orientation differs from $\mathrm{diag}(4, 1)$. A shared center does not imply a shared class shape.

A covariance matrix describes variation and correlation



Same center, different covariance: read the Gaussian shapes

$$\mu = (3,2)^T, \qquad \Sigma_L = \begin{bmatrix} 25 & 0 \\ 0 & 9 \end{bmatrix}, \qquad \Sigma_R = \begin{bmatrix} 15 & 5 \\ 5 & 5 \end{bmatrix}.$$



Compare the two models. Both models center at $(3,2)$. Left: zero covariance and axis-aligned spread. Right: positive covariance and upward tilt.

The axes are features x_1, x_2; colors indicate density height. Moving away from the center along a wide direction is less surprising than moving across a narrow direction.

A correlated Gaussian predicts a missing measurement from the observed one

For a jointly Gaussian feature pair with mean $\mu = (3, 2)^T$ and covariance $\Sigma = \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix}$,

$$X_2 \mid X_1 = x_1 \sim \mathcal{N}\left(\mu_2 + \frac{\Sigma_{21}}{\Sigma_{11}}(x_1 - \mu_1),\ \Sigma_{22} - \frac{\Sigma_{21}^2}{\Sigma_{11}}\right).$$

At $x_1 = 5$, the conditional mean is $2 + (1/4)(2) = 2.5$, and conditional variance is $1 - 1/4 = .75$.

Observing a high first feature raises the expected second feature because covariance is positive. Conditioning also reduces uncertainty. These statements use a joint Gaussian model; a separate diagonal feature model has no such cross-feature adjustment.

Ignoring a missing feature uses a marginal, not a fabricated value

In the preceding Gaussian model, if X_2 is not observed, integrate it out:

$$p(x_1 \mid k) = \int p(x_1, x_2 \mid k) dx_2, \qquad X_1 \mid k \sim \mathcal{N}(\mu_{1k}, \Sigma_{11,k}).$$

Compare two operations. Conditioning on $X_1 = 5$ predicts the distribution of X_2. Marginalizing X_2 uses only the available first-feature likelihood for class comparison.

Setting an unobserved second value to zero would instead claim a specific measurement was observed. Marginalization retains uncertainty about it. The resulting classifier may be less informative because one feature is missing, but its likelihood corresponds to the observation we actually have.

Mahalanobis distance measures deviation in class units

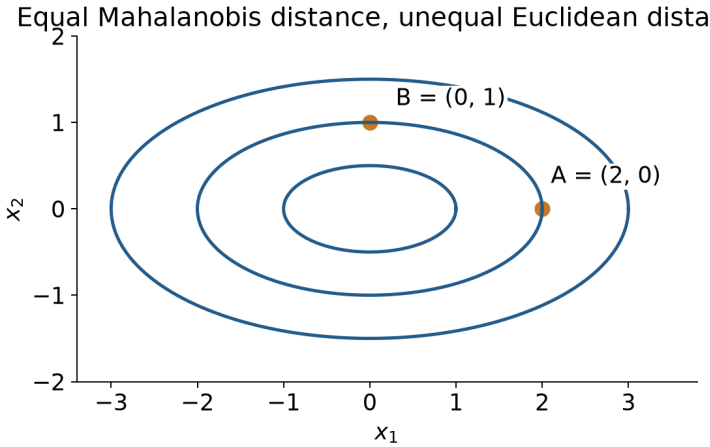
$$d_M^2(x, \mu) = (x - \mu)^T \Sigma^{-1} (x - \mu).$$

For $\mu = 0$ and $\Sigma = \text{diag}(4, 1)$:

Point	Euclidean distance squared	Mahalanobis distance squared	Interpretation
$(2, 0)$	4	$2^2/4 = 1$	One standard deviation in x_1
$(0, 1)$	1	$1^2/1 = 1$	One standard deviation in x_2
$(0, 2)$	4	$2^2/1 = 4$	Two standard deviations in x_2

The broad direction tolerates larger raw deviations. With nonzero covariance, the inverse matrix also accounts for correlated directions. This is a distance relative to a model, not a universal property of a point.

Mahalanobis distance measures deviation in class units



The Gaussian score has three interpretable terms

$$p(x \mid k) = \frac{\exp[-\frac{1}{2} d_{M,k}^2]}{(2\pi)^{d/2} |\Sigma_k|^{1/2}},$$

$$g_k(x) = -\tfrac{1}{2} \log |\Sigma_k| - \tfrac{1}{2} d_{M,k}^2 + \log \pi_k.$$

Score term	Role in comparing classes
$-\frac{1}{2} d_{M,k}^2$	Penalizes departure from this class center and shape
$-\frac{1}{2} \log \lvert\Sigma_k\rvert$	Broad distributions spread their mass over more volume
$\log \pi_k$	Incorporates how common this class is

$-d \log(2\pi)/2$ is common and can be dropped. A class-specific determinant cannot be dropped. Select the largest score under zero-one loss, or normalize and use the loss table.

The multivariate covariance MLE follows from precision and scatter

After fitting the mean, let $S = \sum_i (x_i - \bar{x})(x_i - \bar{x})^T$ and precision $\Lambda = \Sigma^{-1}$.

$$\ell(\Lambda) = \frac{n}{2} \log |\Lambda| - \frac{1}{2} \operatorname{tr}(\Lambda S) + \textit{constant}.$$

Using $\partial \log |\Lambda| / \partial \Lambda = \Lambda^{-T}$ and $\partial \operatorname{tr}(\Lambda S) / \partial \Lambda = S^T$ gives

$$n\Lambda^{-T} = S^T \quad \Longrightarrow \quad \hat{\Sigma} = S/n.$$

For the four centered class-A points, $S = 4I$ and $n = 4$, giving I. The determinant term prevents unlimited spreading, while the distance term measures how the covariance explains observed deviations.

Equal centers still allow a meaningful two-dimensional class decision

Let $x \mid A \sim \mathcal{N}(0, I)$, $x \mid B \sim \mathcal{N}(0, 4I)$, with equal priors. Write $r^2 = x_1^2 + x_2^2$.

$$g_A - g_B = \log 4 - \frac{3}{8} r^2.$$

The boundary satisfies

$$r^2 = \frac{8}{3} \log 4 \approx 3.6968, \qquad r \approx 1.923.$$

At $(1, 0)$, choose the concentrated class A. At $(3, 0)$, choose broad class B. Both classes have center zero, so nearest-center comparison ties everywhere. A's smaller determinant raises central density; B's larger spread penalizes large deviations less. The model compares concentration and shape, not just center location.

Opposite correlations can carry evidence that Gaussian naive Bayes misses

Classes share mean zero and marginal variances 4, but have covariances

$$\Sigma_A = \begin{bmatrix} 4 & 3 \\ 3 & 4 \end{bmatrix}, \qquad \Sigma_B = \begin{bmatrix} 4 & -3 \\ -3 & 4 \end{bmatrix}.$$

At $x = (2, 2)$, determinants are both 7, while $d_A^2 = 8/7$ and $d_B^2 = 8$. With equal priors,

$$g_A - g_B = 24/7, \qquad P(A \mid x) \approx .9686.$$

Gaussian naive Bayes keeps only diagonal variances. It would make the two class models identical and return equal posteriors. The relationship between features, rather than their separate marginal distributions, supplies the class evidence here.

Shared covariance cancels the quadratic terms

If every class shares Σ, expand the distance term:

$$g_k(x) = x^T \Sigma^{-1} \mu_k - \tfrac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k + \text{common terms}.$$

For two classes,

$$g_A(x) - g_B(x) = w^T x + b, \quad w = \Sigma^{-1}(\mu_A - \mu_B),$$
$$b = -\tfrac{1}{2} \mu_A^T \Sigma^{-1} \mu_A + \tfrac{1}{2} \mu_B^T \Sigma^{-1} \mu_B + \log(\pi_A / \pi_B).$$

Why a line? The same $x^T \Sigma^{-1} x$ and log determinant cancel. Different full or diagonal covariances generally leave quadratic terms. “Diagonal” restricts correlations within classes; “shared” restricts differences between classes. These are different assumptions.

Unequal means and variances can create two asymmetric crossings

Take $X \mid A \sim \mathcal{N}(0,1)$ and $X \mid B \sim \mathcal{N}(2,4)$, with equal priors.

$$g_A - g_B = \log 2 - \frac{x^2}{2} + \frac{(x-2)^2}{8}.$$

Setting this difference to zero yields

$$3x^2 + 4x - 4 - 8\log 2 = 0, \qquad x \approx -2.571,\ 1.238.$$

At $x = 0$, A wins; at $x = 2$, B wins; sufficiently negative values also favor broad B. The wider class can occupy both tails even though its center is shifted right. A single midpoint threshold would miss this second crossing because the quadratic terms no longer cancel.

A prior shifts a linear Gaussian threshold

Model. $X \mid A \sim \mathcal{N}(0,1)$ and $X \mid B \sim \mathcal{N}(4,1)$.

$$g_A(x) - g_B(x) = -\tfrac{1}{2}x^2 + \tfrac{1}{2}(x-4)^2 + \log\frac{\pi_A}{\pi_B} = -4x + 8 + \log\frac{\pi_A}{\pi_B}.$$

$$x^* = 2 + \frac{1}{4}\log\frac{\pi_A}{\pi_B}; \qquad \text{choose A if } x < x^*.$$

Priors	Threshold	Interpretation
$(.5, .5)$	2	Midpoint between centers
$(.8, .2)$	2.347	More common A receives a larger region

The class means and variances did not move. The boundary changed because the prior preference changed. Unequal losses can shift the final action threshold further.

Unequal variances can produce two thresholds

Model. $X \mid A \sim \mathcal{N}(0,1)$, $X \mid B \sim \mathcal{N}(0,4)$; equal priors.

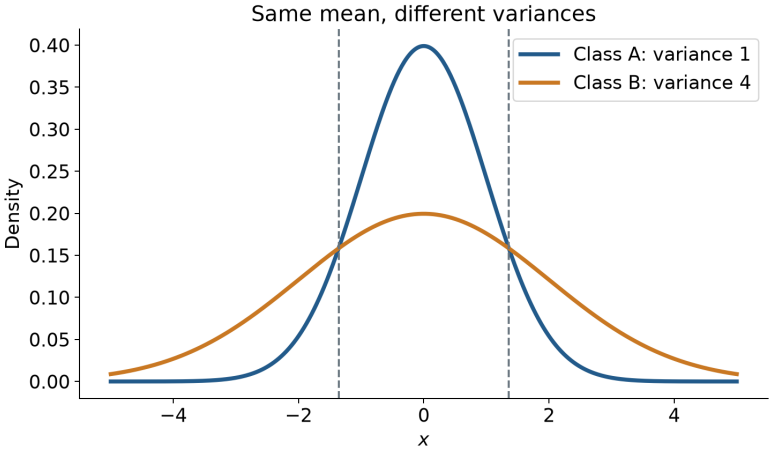
$$g_A(x) - g_B(x) = \log 2 - \frac{3}{8}x^2.$$

$$g_A = g_B \quad \Longrightarrow \quad |x| = \sqrt{\frac{8}{3}\log 2} \approx 1.360.$$

Input region	Winning class	Explanation
$\lvert x\rvert < 1.360$	Narrow class A	More concentrated near the common center
$\lvert x\rvert > 1.360$	Broad class B	More support in either tail

The determinant favors the narrow class near the center; the distance penalty favors the broad class far away. A nearest-center rule cannot distinguish these classes because their means are identical.

Unequal variances can produce two thresholds



Costs shift a fitted Gaussian decision boundary

Keep the eight-point fitted model, equal priors, and shared $\Sigma = I$. Let A mean sound and B mean defective. Its log posterior odds are

$$\log \frac{P(B \mid x)}{P(A \mid x)} = 4(x_1 + x_2 - 6).$$

With miss cost 30 and false-alarm cost 2, select the defect action when the posterior odds exceed $2/30$ (equivalently, log odds exceed $\log(2/30)$):

$$x_1 + x_2 > 6 + \tfrac{1}{4} \log(2/30) \approx 5.323.$$

At $(2.8, 2.8)$, $P(B \mid x) \approx .168$. Sound is the most likely class, but the defect action minimizes risk. The covariance and means remain fitted measurements; costs change the final intercept.

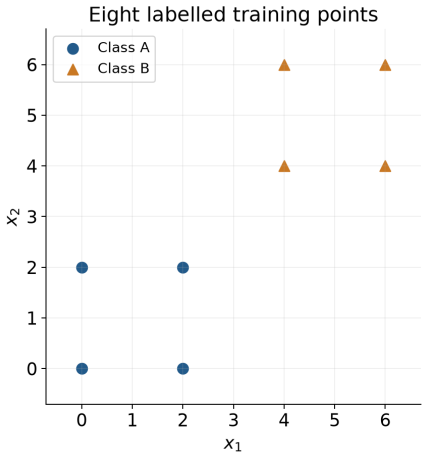
Eight points let us train the whole model by hand

Class	Labelled feature vectors	Class mean
A	$(0,0), (2,0), (2,2), (0,2)$	$(1,1)^T$
B	$(4,4), (6,4), (6,6), (4,6)$	$(5,5)^T$

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i, \qquad \hat{\pi}_k = \frac{n_k}{n}.$$

Training quantities. Both classes contain 4 points, so both priors are .5. **What remains?** The shape around each center. Compute deviations from the **class** mean, not the grand mean $(3,3)^T$. Rows in the implementation will be samples; columns will be features. This convention determines which axis computes the means.

Eight points let us train the whole model by hand



Covariance is an average of centered outer products

$$\hat{\Sigma}_k = \frac{1}{n_k} \sum_{i:y_i=k} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T.$$

A centered vector	δ_1^2	$\delta_1\delta_2$	δ_2^2
$(-1,-1)$	1	1	1
$(1,-1)$	1	-1	1
$(1,1)$	1	1	1
$(-1,1)$	1	-1	1
Sum	4	0	4

$$\hat{\Sigma}_A = \hat{\Sigma}_B = I.$$

MLE divides by $n_k = 4$. The unbiased sample covariance divides by $n_k - 1 = 3$ and gives $\frac{4}{3}I$; they are different estimators. An explicit denominator makes the derivation and code agree.

Update sufficient statistics as new samples arrive

For each class, keep count n, mean μ_n, and scatter M_n:

$$\delta = x_{n+1} - \mu_n, \quad \mu_{n+1} = \mu_n + \frac{\delta}{n+1},$$

$$M_{n+1} = M_n + \delta(x_{n+1} - \mu_{n+1})^T, \qquad \hat{\Sigma}_n = M_n/n.$$

Scalar samples seen	Count	Mean	Scatter
2	1	2	0
2, 4	2	3	2
2, 4, 6	3	4	8

The final MLE variance is $8/3$, matching batch centering. Maintain separate statistics for each labelled class. Updates add evidence; they do not require retaining every past observation or treating a new prediction as a known training label.

Pool within classes, then derive the boundary

$$\hat{\Sigma}_{shared} = \frac{1}{n} \sum_{k} \sum_{i:y_i=k} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T = I.$$

Centering all eight points by the grand mean instead gives

$$\mathrm{Cov}_{all}(X) = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix},$$

which contains between-class separation and is not the required within-class covariance. With equal priors and $\Sigma = I$, compare squared distances:

$$\|x - (1,1)^T\|^2 = \|x - (5,5)^T\|^2 \quad \Longrightarrow \quad x_1 + x_2 = 6.$$

Choose A for $x_1 + x_2 < 6$ and B for $x_1 + x_2 > 6$. The fitted boundary follows from the fitted parameters; it is not drawn independently.

Scores, posterior, and boundary should agree

Fitted model: $\mu_A = (1,1)^T$, $\mu_B = (5,5)^T$, $\Sigma = I$, equal priors.

New point	d_A^2	d_B^2	$g_A - g_B$
$(2,3)$	5	13	4
$(4,3)$	13	5	-4

$$P(A \mid x) = \frac{1}{1 + \exp[-(g_A - g_B)]}.$$

Thus $P(A \mid (2,3)) \approx .982$ and $P(A \mid (4,3)) \approx .018$. **Geometry check.** The coordinate sums are 5 and 7, on opposite sides of $x_1 + x_2 = 6$. **Action check.** A cost-sensitive decision still requires a loss table; a confident class score alone does not specify the cheapest action.

Implement the same computation without matrix inversion

```
def score(x, mu, Sigma, prior):
    delta = x-mu
    np.linalg.cholesky(Sigma)  # requires positive definiteness
    _, logdet = np.linalg.slogdet(Sigma)
    d2 = delta @ np.linalg.solve(Sigma, delta)
    return -.5*(logdet+d2) + np.log(prior)
means = [np.array([1.,1.]), np.array([5.,5.])]
s = np.array([score(np.array([4.,3.]), m, np.eye(2), .5)
              for m in means])
q = np.exp(s-s.max())
print(q/q.sum())  # approximately [.018, .982]
```

Solve $\Sigma v = \delta$, then compute $\delta^T v$. Check class order, covariance definiteness, and normalization. The numerical routine returns exactly the log score we derived, with the shared normalizer omitted.

A covariance structure changes the amount of information required

For 3 classes and $d = 20$, count covariance entries:

Structure	Covariance parameters	What is shared or omitted?
Separate full	$3(210) = 630$	No class sharing
Shared full	210	One common class shape
Separate diagonal	60	No within-class correlations
Shared spherical	1	One variance for every direction

Means add 60 parameters and priors add 2. With only 10 labelled examples per class, each full empirical covariance has rank at most 9. Flexible parameterization does not guarantee estimability; class sharing and restrictions trade expressive shape information for stable learning.

Small samples can leave covariance directions unlearned

For collinear points $(0,0),(1,2),(2,4)$,

$$\hat{\Sigma} = \begin{bmatrix} 2/3 & 4/3 \\ 4/3 & 8/3 \end{bmatrix}, \qquad |\hat{\Sigma}| = 0.$$

The samples vary in only one direction. In general,

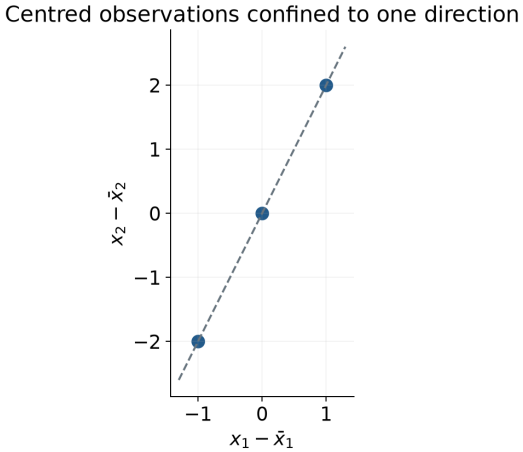
$$\mathrm{rank}(\hat{\Sigma}_k) \le \min(d, n_k - 1).$$

Shrinkage.

$$\tilde{\Sigma} = (1-\lambda)S + \lambda\tau I, \qquad \tau = \mathrm{tr}(S)/d.$$

At $\lambda = .2$, this example's eigenvalues change from $(0, 10/3)$ to $(1/3, 3)$. Full class covariances learn $Kd(d+1)/2$ entries; shared, diagonal, or spherical structures learn fewer. Choosing a simpler shape exchanges flexibility for estimation stability. Select the amount of shrinkage with validation data.

Small samples can leave covariance directions unlearned



Shrinkage places a floor under poorly measured directions

An empirical covariance has eigenvalues $(0, .01, 5)$. Its spherical target variance is

$$\tau = \operatorname{tr}(S)/3 = 1.67.$$

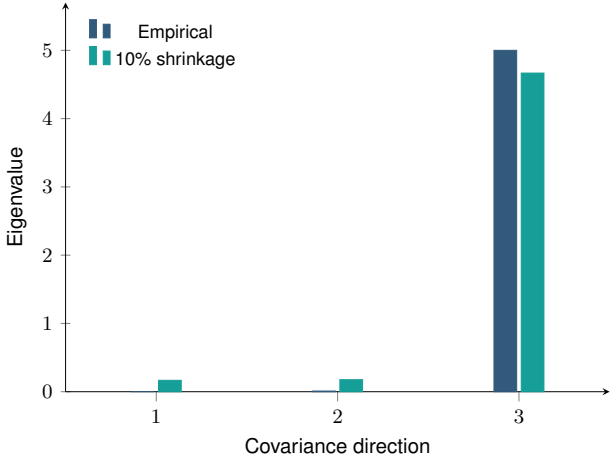
For $\tilde{\Sigma} = .9S + .1\tau I$, the new eigenvalues are

$$(.167,\ .176,\ 4.667).$$

The condition number changes from infinite to about $4.667/.167 \approx 27.95$. A direction with almost no measured variation no longer receives an unlimited inverse weight.

This stabilizes computation but introduces a shape assumption. A very large shrinkage coefficient can erase real anisotropy. The amount should be chosen using separate validation evidence, while the final test data remain for evaluation.

Shrinkage places a floor under poorly measured directions



Exercise: a very confident class may still be the costly action

Let $\mu_A = (0, 0)$, $\mu_B = (3, 0)$, shared covariance $\mathrm{diag}(1, 4)$, and priors $(.75, .25)$. Observe $x = (0, 2)$.

1. Compute both Mahalanobis squared distances.
2. Use the common determinant to simplify the log-score difference.
3. Calculate the class posterior.
4. Choosing A when truth is B costs 1,000; choosing B when truth is A costs 1. Compare action risks.

The large cost is an explicit teaching assumption. Keep the state posterior separate from the action choice. Explain why the second coordinate gives the same distance contribution to both classes in this model.

Solution: compare the residual probability with the loss ratio

Distances are $d_A^2 = 1$ and $d_B^2 = 10$. Thus

$$g_A - g_B = \frac{10 - 1}{2} + \log 3 = 4.5 + \log 3 \approx 5.5986,$$

$$P(A \mid x) \approx .99631, \qquad P(B \mid x) \approx .00369.$$

Choosing A risks about $1,000(.00369) = 3.69$; choosing B risks about $1(.99631) = .99631$. Select B under the stated costs.

The posterior favors A strongly. The very expensive residual error makes B the cheaper action. Equal costs would instead select A. Probability, covariance geometry, prior preference, and loss each play a distinct role in the same calculation.

Final exercise: change the prior and the action

Keep the fitted eight-point means and $\Sigma = I$. In a new population, priors are $(\pi_A, \pi_B) = (.8, .2)$. Observe $x = (3, 3)^T$.

1. Calculate the two distances. Which part of the score decides between classes?
2. Compute $P(A \mid x)$ and $P(B \mid x)$.
3. Choosing A when truth is B costs 10. Choosing B when truth is A costs 1. Correct decisions cost 0. Choose the minimum-risk action.

Extension. Explain why replacing $\Sigma = I$ by two different diagonal matrices need not preserve a linear boundary. State exactly which equality allowed the quadratic terms to cancel.

Solution and the bridge to discriminative classifiers

$$d_A^2 = d_B^2 = 8, \qquad P(A \mid x) = .8, \quad P(B \mid x) = .2.$$

Action	Risk	Decision
Choose A	$10(.2) = 2$	Higher loss
Choose B	$1(.8) = .8$	Minimum risk

What we can now explain.

- Counts estimate priors and class-conditional parameters.
- Bayes combines competing explanations into a posterior.
- Loss turns a posterior into an action; it can reverse the largest-posterior decision.
- Class shape assumptions determine whether Gaussian boundaries are linear or quadratic.

Next lecture: study the boundary directly, then find a projection that separates classes relative to their within-class spread.